Suriya M.J

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EDUCATION

National Institute of Technology, Warangal
B.Tech – Chemical Engineering | 2024-2028 | CGPA:6.53

New Millennium School, Bahrain
Class XII | 2024 | 85.6%
Class X | 2022 | 93%

PROJECTS

Cyclone Separator Performance Analysis – Lapple Model *(March 2026)*

- Modelled grade efficiency and cut size of a standard Lapple cyclone separator using Microsoft Excel
- Performed sensitivity analysis varying inlet velocity from 10-25m/s, demonstrating 37% reduction in cut size and nearly doubling collection efficiency at 5µm
- Calculated pressure drop of 2160 Pa using Shepherd & Lapple correlation

Drug Dissolution Rate Analysis – Noyes-Whitney Model *(March 2026)*

- Analysed pharmaceutical dissolution kinetics of paracetamol using Noyes-Whitney model in Microsoft Excel
- Developed Python scripts using matplotlib to generate dissolution profiles and sensitivity analysis graphs
- Demonstrated nearly 5x improvement in dissolution by increasing particle surface area from 0.0005 to 0.005m²

Design Thinking Case Study – Tic Tac Packaging Redesign *(November 2024)*

- Identified usability limitation in Tic Tac packaging and proposed redesigned single-dispense mechanism
- Awarded S grade (100/100) for communication and conceptual thinking

TECHNICAL SKILLS

Tools & Software: Microsoft Excel, Python (matplotlib, NumPy), Audacity, Canva, MS Word

Programming: Python – data analysis, numerical computation, visualization | C++ | MySQL – database management

Web Development: HTML, CSS

Languages Known: English, Hindi, Tamil, French

LEADERSHIP & EXPERIENCE

President – Rising Stars Gavel Club *(July 2020-June2021)*

- Led club operations during COVID-19, managing virtual meetings and sustaining member engagement
- Won 50+ awards including Best Speaker, Best Evaluator and Table Topics across multiple Sessions
- Recognized for confident stage presence, vocal clarity and audience engagement

Additional Secretary – International Students Working Group(ISWG), NIT Warangal *(August2025-Present)*

- Representing International Students in official working group at NIT Warangal
- Served as Master of Ceremonies for official ISWG events

Master of Ceremonies

- Hosted and moderated multiple official events at school and college level
- Strong public speaking, time management and event coordination skills

MEMBERSHIPS

Student member , Indian Institute of Chemical Engineering (IIChE)

ADDITIONAL INFORMATION

Academic Interests: Pharmaceutical engineering, process design, separation processes, oil & gas, sustainable chemical processes, heat transfer applications, fluid mechanics, and industrial chemical manufacturing.

Portfolio: suriyamj.in

EXTRACURRICULAR & INTERESTS

- Multilingual communicator – fluent in Tamil, English, Hindi and French, with international school background in Bharain
- Passionate about pharmaceutical engineering and process design with focus on bridging chemical engineering fundamentals with industry applications.